

Converting Between Exponential and Logarithmic Forms

Answer Key

Algebra 2 Exponential and Logarithmic Functions

Quick Answers

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|----------|--------------------------------------|
| 1. 3 | 6. $\frac{1}{3}$ |
| 2. 32 | 7. 8 |
| 3. 4 | 8. $\frac{\log(x)}{\log(3)}$, 2.727 |
| 4. -2 | 9. 14 |
| 5. 5.322 | 10. 15.747 |
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Common wrong answers

5. *If they got 20: divided 40 by 2 instead of taking a logarithm*
8. *If they got 1.89712: computed the natural log of the quotient instead of the quotient of natural logs*
10. *If they got 1.91388: divided the doubling factor by the growth factor instead of converting to a logarithm*
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1. $5^3 = 125$ into logarithmic form.

Step 1. Label the three roles before moving anything: the base is 5, the exponent is 3, and the result is 125.

Step 2. The base stays the base and the exponent becomes what the logarithm equals, so $5^3 = 125$ becomes $\log_5 125 = 3$. 3

Say it in words: “the power of 5 that gives 125 is 3.” Reading a logarithm as a question about an exponent is what makes both forms obvious.

2. $\log_2 32 = 5$ into exponential form.

Step 1. In $\log_b x = y$ the subscript is the base, so here $b = 2$, $x = 32$, and the value of the logarithm, 5, is the exponent.

Step 2. Rebuild the power: $2^5 = 32$, and evaluating it confirms the statement. 32

Common slip: writing 5^2 instead of 2^5 . The subscript is always the base — it never becomes the exponent.

3. Evaluate $\log_3 81$.

Step 1. A logarithm with a nice base is best answered from a short table of powers rather than a calculator: $3^1 = 3$, $3^2 = 9$, $3^3 = 27$, $3^4 = 81$.

Step 2. $\log_3 81$ asks which exponent produces 81, and the table gives $3^4 = 81$. 4

Both forms: $3^4 = 81 \iff \log_3 81 = 4$ — the same three numbers, arranged for a different question.

4. $10^{-2} = 0.01$ into logarithmic form.

Step 1. The roles do not change just because the exponent is negative: base 10, exponent -2 , result 0.01.

Step 2. So $\log_{10} 0.01 = -2$.

-2

Part (c): raising a base bigger than 1 to a positive power gives a result bigger than 1, so any result between 0 and 1 must have come from a negative exponent. The sign of the logarithm reports which side of 1 the result is on.

5. Solve $2^x = 40$.

Step 1. The unknown is the exponent, and logarithmic form is the one that puts the exponent alone on a side: $x = \log_2 40$.

Step 2. Part (b): $2^5 = 32$ and $2^6 = 64$, and 40 lies between them, so x is between 5 and 6.

Step 3. By change of base, $x = \frac{\ln 40}{\ln 2} = 5.32193\dots$, which rounds to three decimals.

5.322

Watch out: dividing 40 by 2 gives 20 and answers a different question entirely — that is the exponent's job, not a factor's. The estimate in part (b) is the cheapest protection against it.

6. Evaluate $\log_8 2$ exactly.

Step 1. Convert first: $\log_8 2 = y$ means $8^y = 2$, so the question is which power of 8 gives 2.

Step 2. Since $8 = 2^3$, we need $2^{3y} = 2^1$, so $3y = 1$.

$\frac{1}{3}$

Sanity check on size: the result 2 is smaller than the base 8 but bigger than 1, so the exponent had to land strictly between 0 and 1 — and $\frac{1}{3}$ does.

7. Solve $\log_4 x = \frac{3}{2}$.

Step 1. The unknown is inside the logarithm, so exponential form is the one that frees it: $x = 4^{3/2}$.

Step 2. A fractional exponent is a root and a power together: the denominator 2 says square root, the numerator 3 says cube it, so $x = (\sqrt{4})^3 = 2^3$.

8

Check (part c): $\log_4 8 = \frac{3}{2}$ because $4^{3/2} = 8$, so the value satisfies the original statement.

8. Change of base for $\log_3 x$, then $\log_3 20$.

Step 1. Start from the conversion: $\log_3 x = y$ means $3^y = x$. Taking natural logarithms of both sides gives $y \ln 3 = \ln x$.

Step 2. Dividing by $\ln 3$ isolates y , which is the change-of-base formula.

$\log_3 x = \frac{\ln x}{\ln 3}$

Step 3. Part (b): $\log_3 20 = \frac{\ln 20}{\ln 3} = \frac{2.995732}{1.098612} = 2.72683\dots$

2.727

Watch out: $\ln\left(\frac{20}{3}\right) = 1.897$ is not $\frac{\ln 20}{\ln 3}$. The formula divides two logarithms; it does not take the logarithm of a quotient.

9. Solve $\log_5(2x - 3) = 2$.

Step 1. Whatever the logarithm is taken of becomes the result in exponential form, brackets and all: $2x - 3 = 5^2 = 25$.

Step 2. Now it is an ordinary linear equation: $2x = 28$.

$$\boxed{x = 14}$$

Part (c): substituting gives $2(14) - 3 = 25$, which is positive, as it must be — a logarithm is only defined on positive inputs, so any candidate that made $2x - 3$ zero or negative would have to be rejected as extraneous.

10. Challenge: doubling time at 4.5 percent a year.

Step 1. Part (a): doubled means the value equals twice the starting amount, so $1200 \cdot 1.045^t = 2400$.

Step 2. Part (b): dividing both sides by the starting amount leaves $1.045^t = 2$ — notice the starting amount cancels, so the doubling time does not depend on it. The unknown is an exponent, so convert: $t = \log_{1.045} 2$.

Step 3. Change of base gives $t = \frac{\ln 2}{\ln 1.045} = \frac{0.693147}{0.044017} = 15.74731 \dots$ years.

$$\boxed{15.747}$$

Watch out: $\frac{2}{1.045} = 1.914$ answers “how many times bigger,” not “after how long.” Whenever the unknown is the exponent, the conversion to logarithmic form is the whole move.